# CSS Cheatsheet

## Font

### Font-Family

```
font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; /* Sets prioritized fonts for text */
```

### Font-Style

```
font-style: italic; /* Makes text italic */
```

### Font-Variant

```
font-variant: small-caps; /* Displays lowercase as small capital letters */
```

### Font-Weight

```
font-weight: bold; /* or 100-900 for thickness control */
```

### Font-Size

```
font-size: 1rem; /* Sets text size, responsive with rem */
```

### Font (Shorthand)

```
font: italic small-caps bold 1rem/1.5 'Segoe UI'; /* Combines style, variant, weight, size, line-height, family */
```

### Font-Kerning

```
font-kerning: normal; /* Controls spacing between letters */
```

### Font-Stretch

```
font-stretch: condensed; /* Expands or condenses text width */
```

### Font-Variant-Ligatures

```
font-variant-ligatures: common-ligatures; /* Enables typographic ligatures */
```

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## Text

### Text-Align

```
text-align: justify; /* Aligns text evenly across the line */
```

### Letter-Spacing

```
letter-spacing: 0.15em; /* Adjusts space between characters */
```

## Text-Decoration

```
text-decoration: underline; /* Adds decoration like underline/overline/line-through */
```

## Word-Spacing

```
word-spacing: 0.25em; /* Sets spacing between words */
```

## Text-Transform

```
text-transform: uppercase; /* Converts text to uppercase */
```

## Text-Indent

```
text-indent: 0.5cm; /* Indents the first line of text */
```

## Line-Height

```
line-height: 1.5; /* Sets vertical spacing between lines */
```

## Text-Overflow

```
text-overflow: ellipsis; /* Adds ... when text overflows */
```

## White-Space

```
white-space: nowrap; /* Prevents text from wrapping */
```

### Word-Break

```
word-break: break-word; /* Breaks long words to fit */
```

### Hyphens

```
hyphens: auto; /* Allows automatic hyphenation */
```

---

## Background

### Background-Image

```
background-image: url("path"), linear-gradient(...); /* Adds image or gradient */
```

### Background-Position

```
background-position: center; /* Positions background image */
```

### Background-Size

```
background-size: cover; /* Scales image to cover element */
```

### Background-Clip

```
background-clip: text; /* Clips background to text */
```

### Background-Blend-Mode

```
background-blend-mode: multiply; /* Blends background layers */
```

### Background (Shorthand)

```
background: url('image.jpg') no-repeat center/cover; /* Sets multiple background properties at once */
```

---

## Border & Outline

### Border

```
border: 2px solid #000; /* Adds border with size, style, color */
```

### Border-Radius

```
border-radius: 15px; /* Rounds corners */
```

### Outline

```
outline: 2px dashed red; /* Adds outline outside element box */
outline-offset: 4px; /* Adjusts distance from element */
```

### Border-Image

```
border-image: url(border.png) 30 round; /* Uses image as border */
```

---

# Box Model

## Margin & Padding

```
margin: 10px 20px; /* Outer spacing */
padding: 5px 10px; /* Inner spacing */
```

## Width & Height

```
width: 100%; /* Element width */
height: fit-content; /* Height based on content */
```

## Box-Sizing

```
box-sizing: border-box; /* Includes padding/border in total size */
```

## Overflow

```
overflow: hidden; /* Hides overflow */
overflow-x: scroll; /* Scroll horizontally */
overflow-y: auto; /* Scroll vertically as needed */
```

## Aspect-Ratio

```
aspect-ratio: 16/9; /* Keeps width:height ratio */
```

## Object-Fit

```
object-fit: cover; /* Scales image/video to fill box */
```

# Colors

## Color

```
color: #333; /* Text color */
```

## Opacity

```
opacity: 0.8; /* Element transparency */
```

## Modern Colors

```
color: hsl(200, 80%, 40%); /* HSL color format */
```

---

# Positioning

## Position

```
position: relative; /* Sets positioning context */
```

## Z-Index

```
z-index: 10; /* Stacking order */
```

---

# Flexbox

## Parent (Container)

```
display: flex; /* Activates flex layout */
flex-direction: row; /* Row direction */
justify-content: space-between; /* Distributes space */
align-items: center; /* Aligns items vertically */
gap: 1rem; /* Space between flex items */
```

## Child (Item)

```
flex: 1; /* Flexible width */
align-self: center; /* Individual alignment */
order: 2; /* Changes visual order */
```

---

# CSS Grid

## Parent (Container)

```
display: grid; /* Enables grid layout */
grid-template-columns: 1fr 2fr; /* Defines columns */
grid-template-rows: auto 1fr auto; /* Defines rows */
grid-template-areas:
  "header header"
  "sidebar main"
  "footer footer"; /* Named grid areas */
gap: 1rem; /* Space between grid items */
```

## Child (Item)

```
grid-area: main; /* Places item in named area */
```

## CSS Variables

```
:root {
  --primary-color: #007bff; /* Declares variable */
}

div {
  color: var(--primary-color); /* Uses variable */
}
```

## Media Queries

```
@media (max-width: 768px) {
  body {
    font-size: 14px; /* Applies styles on small screens */
  }
}
```

## Animations & Transitions

### Animation

```
@keyframes slide {
  from { transform: translateX(-100%); }
  to { transform: translateX(0); }
}

.element {
  animation: slide 1s ease-in-out infinite alternate; /* Animates element */
}
```

### Transition

```
transition: all 0.3s ease-in-out; /* Smoothly animates property changes */
```

### Transform

```
transform: scale(1.1) rotate(10deg); /* Moves, rotates, scales elements */
```

---

## Modern Features

### Scroll Snap

```
scroll-snap-type: x mandatory; /* Creates snap scrolling */
```

### Backdrop Filter

```
backdrop-filter: blur(10px); /* Adds blur behind element */
```

### Custom Scrollbar

```
::-webkit-scrollbar { width: 8px; }
::-webkit-scrollbar-thumb { background: #888; border-radius: 4px; } /* Styles scrollbar */
```

### Responsive Sizing

```
font-size: clamp(1rem, 2vw, 1.5rem); /* Responsive font size */
```

# CSS

## Selectors

```css
/* Basic */
p { color: red; }
#id { font-size: 18px; }
.class { background: yellow; }

/* Attribute */
input[type="text"] { border: 1px solid black; }

/* Descendant & Child */
div p { color: gray; }      /* inside div */
div > p { color: blue; }    /* direct child */

/* Sibling */
h1 + p { margin-top: 0; }   /* immediate */
h1 ~ p { color: green; }    /* all following */

/* Pseudo-class */
a:hover { color: tomato; }
input:focus { border-color: dodgerblue; }
li:nth-child(2) { color: purple; }

/* Pseudo-element */
p::first-line { font-weight: bold; }
p::before { content: "→ "; }
```

## Box Model

Every element = box with:

```
margin → border → padding → content
```

```
div {
  margin: 10px;      /* outside space */
  border: 2px solid black;
  padding: 10px;     /* inside space */
  width: 200px;
  height: 100px;
  box-sizing: border-box; /* include padding/border in total size */
}
```

## Display & Position

```
/* Display */
div { display: block; }
span { display: inline; }
img { display: inline-block; }
.hidden { display: none; }

/* Position */
.box {
  position: relative; /* reference point for absolute children */
  top: 10px; left: 20px;
}
.absolute {
  position: absolute; /* relative to nearest positioned ancestor */
  top: 0; right: 0;
}
.fixed { position: fixed; bottom: 10px; right: 10px; }
.sticky { position: sticky; top: 0; }
```

## Flexbox (for sane layouting)

```css
.container {
  display: flex;
  justify-content: center;   /* horizontal */
  align-items: center;       /* vertical */
  flex-direction: row;       /* or column */
  flex-wrap: wrap;
  gap: 10px;
}

.item {
  flex: 1; /* fills remaining space */
}
```

## Grid (for the perfectionists)

```css
.container {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  grid-template-rows: auto;
  gap: 10px;
}

.item1 {
  grid-column: 1 / 3; /* spans 2 columns */
}
```

## Typography

```css
body {
  font-family: 'Poppins', sans-serif;
  font-size: 16px;
  line-height: 1.5;
```

```
  color: #222;
}

h1 { text-transform: uppercase; letter-spacing: 2px; }
a { text-decoration: none; color: royalblue; }
```

## Colors & Backgrounds

```
div {
  color: #333;
  background-color: lightblue;
  background-image: url('img.jpg');
  background-size: cover;
  background-position: center;
}
```

## Borders, Shadows, Corners

```
.box {
  border: 2px solid #333;
  border-radius: 10px;
  box-shadow: 0 4px 6px rgba(0,0,0,0.1);
}
```

## Transitions & Animations

```
button {
  background: crimson;
  transition: background 0.3s ease;
}

button:hover {
```

```
  background: darkred;
}

/* Keyframes */
@keyframes fadeIn {
  from { opacity: 0; }
  to { opacity: 1; }
}
div { animation: fadeIn 1s ease-in; }
```

## Responsive Design

```
@media (max-width: 768px) {
  body { font-size: 14px; }
  .container { flex-direction: column; }
}
```

## Variables (Custom Properties)

```
:root {
  --main-color: #3498db;
  --radius: 8px;
}

button {
  background: var(--main-color);
  border-radius: var(--radius);
}
```

## Z-Index Logic

Higher `z-index` = above others, but only if element is positioned.

```
.modal { position: fixed; z-index: 9999; }
```

## Common Tricks

```
/* Center anything with Flex */
.center {
  display: flex;
  justify-content: center;
  align-items: center;
}

/* Hide scrollbar */
::-webkit-scrollbar { display: none; }

/* Responsive image */
img { max-width: 100%; height: auto; }

/* Text ellipsis */
.text {
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
}
```